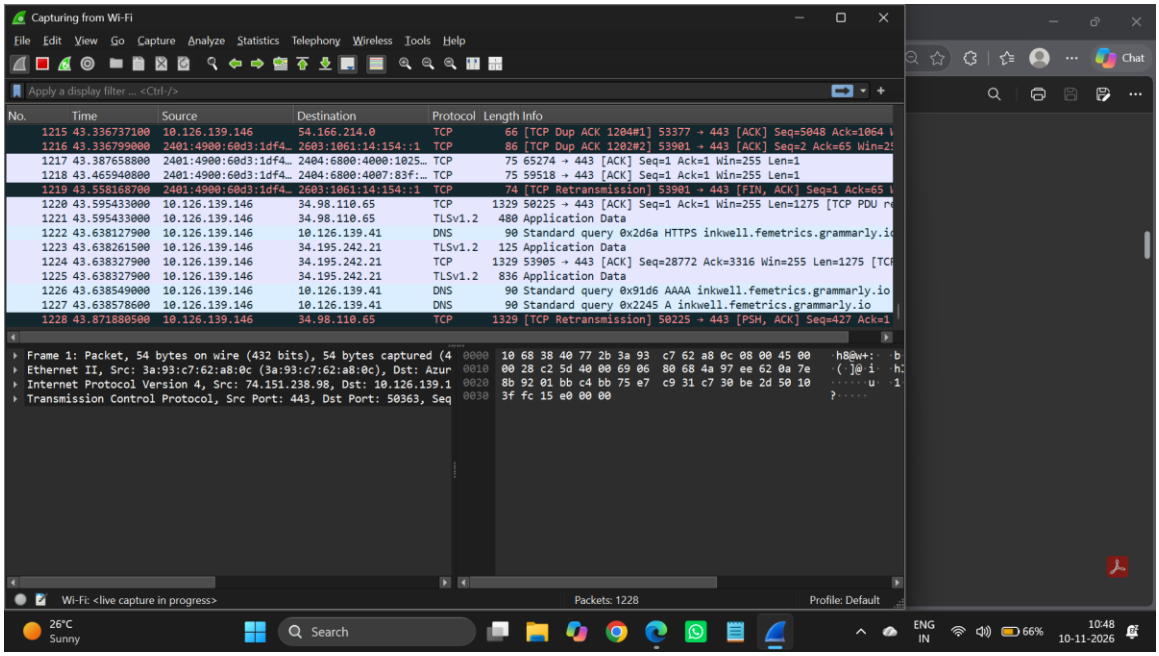
SKILL 1

Roll No. : 2520030072

Name: Manasvi Param

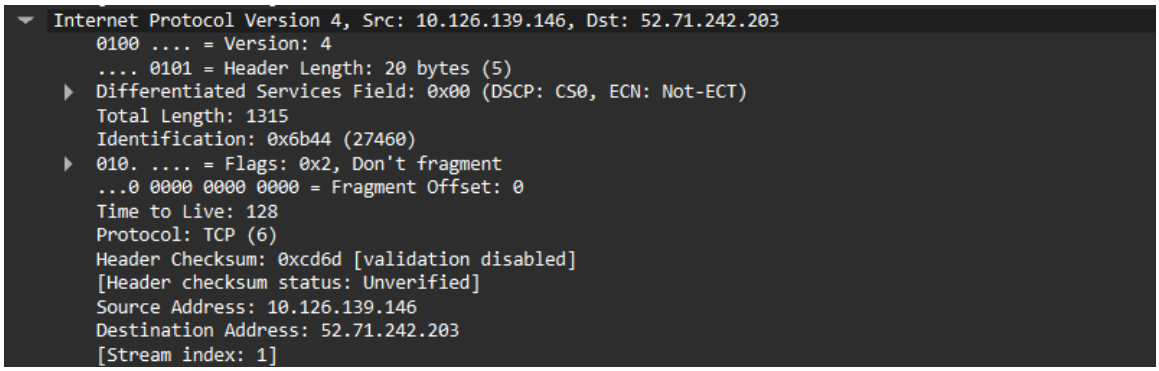
Wireshark Packet Capture

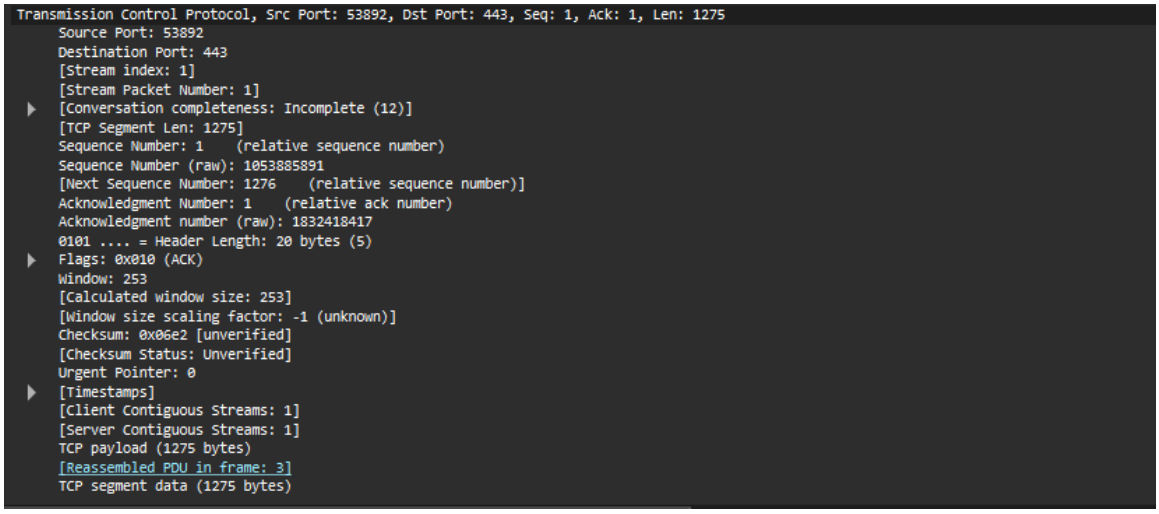
Wireshark was used to capture packets over the Wi-Fi interface after generating network traffic by opening websites.



TCP Packet Header Details

Selected packet: Packet No. 2. The packet has a frame length of 1329 bytes.

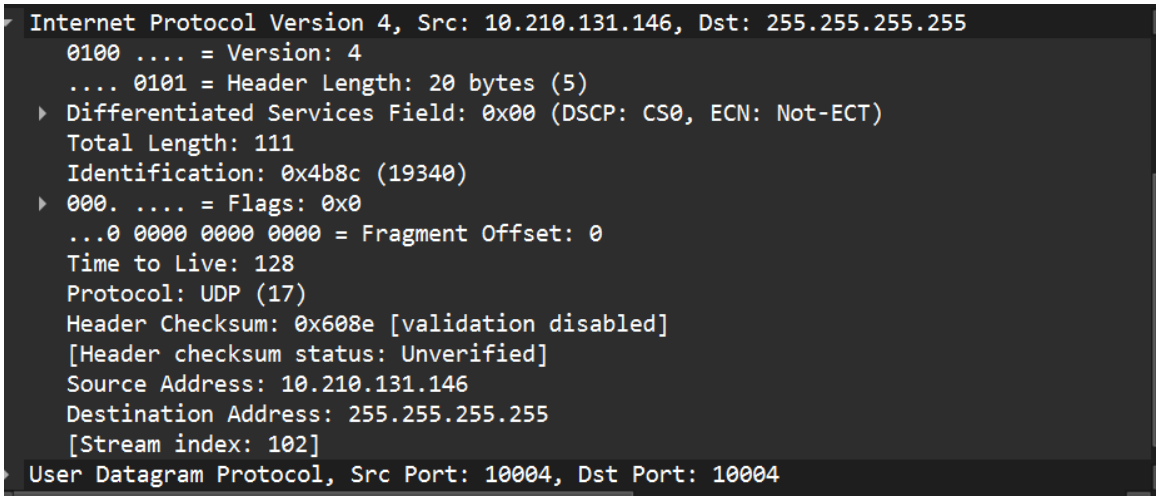


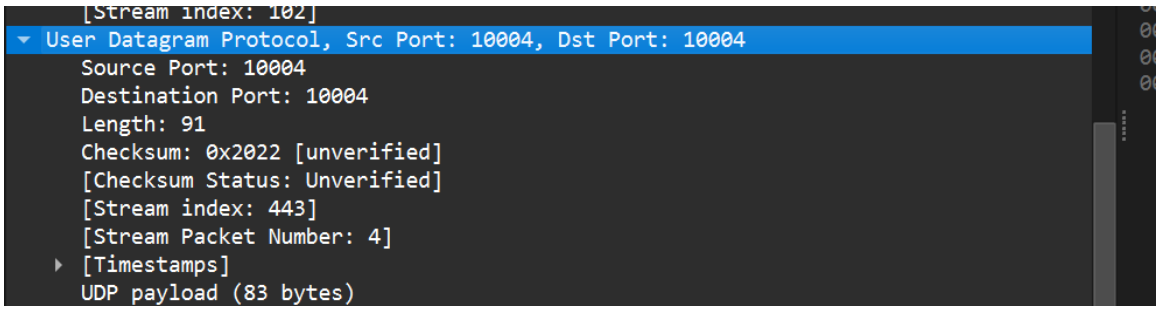


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UDP Packet Header Details

Selected packet: Packet No. 45839. The packet has a frame length of 125 bytes.





Observation Table

Parameter	TCP Packet (No. 2)	UDP Packet (No. 45839)
Frame Length	1329 bytes	125 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes (TCP)	8 bytes (UDP)
IP Total Length	1315 bytes	111 bytes

Transport Length	1295 bytes	91 bytes
Payload Length	1275 bytes	83 bytes

Header Overhead and Payload Efficiency

Calculation	TCP	UDP
Header overhead	14 + 20 + 20 = 54 bytes	14 + 20 + 8 = 42 bytes
Payload efficiency	(1275 / 1329) × 100 = 95.94%	(83 / 125) × 100 = 66.40%

The calculations use the captured frame lengths shown in Wireshark. The Ethernet FCS was not included in these captured frame lengths.

Answers

1. What is encapsulation?

Encapsulation is the process of adding protocol information to data as it moves through the network layers. For example, TCP adds its header, then IP adds another header, and Ethernet adds its header before the data is sent.

2. What is the purpose of the Ethernet header?

The Ethernet header helps deliver a frame across a local network. It mainly contains the source and destination MAC addresses and identifies the type of data carried inside the frame.

3. What information is stored in the IP header?

The IP header contains details such as the source and destination IP addresses, IP version, packet length and protocol type. Routers use this information to forward packets to the right destination.

4. What is the difference between TCP and UDP headers?

A TCP header is larger and contains information used for reliable, ordered communication, such as sequence and acknowledgement numbers. A UDP header is simpler and contains source port, destination port, length and checksum fields. A basic TCP header is 20 bytes, while a UDP header is 8 bytes.

5. Why does each protocol add a header?

Each protocol adds a header because it has a different job. The header carries the information needed for that job, such as identifying devices, routing packets or managing communication between applications.

6. What is packet overhead?

Packet overhead is the extra information added to a packet for delivery and control. It includes protocol headers and, when present, trailers such as the Ethernet FCS. It is separate from the actual application data.

7. How does header overhead affect network efficiency?

Headers take up space in a packet, leaving less room for application data. If the headers make up a large part of the packet, a smaller percentage of the transmitted bytes carries useful data, so efficiency goes down.